Assignment#1: Probability Models

General Instructions:

- There are two problems. Carefully read all notes and questions.
- Answers should be clearly typed as in a technical report
- Type out the whole solutions and answers, submit a PDF report on Canvas (or other associated material)
- Scores for rubrics are shown in [ ] within each question

Problem 1: Naive Bayes for Customer Churn Using Observed Data [50 pts]

Background

You are a data scientist working for a subscription-based company. Your goal is to predict whether a customer will churn (Y=1) or not churn (Y=0) using behavioral data.

You will build two Naive Bayes models, each using a different feature set. For each model, you will:

1. Estimate probabilities from data
2. Apply Laplace smoothing
3. Compute posterior churn probabilities
4. Compare model conclusions

Requirements

- Use Naive Bayes classification
- Derive all probabilities from the provided datasets
- Use Laplace ($\alpha=1$) smoothing for all conditional probabilities
- Compute normalized posterior probabilities
- Show all calculations clearly
- Use a calculator for computing, do not use machine learning libraries

Given (common to both models)

Let Y represent the event churn. From historical data, the overall churn rate is:

- $P(Y = 1) = 0.3$
- $P(Y = 0) = 0.7$

Model A — Support-related features

Feature definitions

- ($X_1 = 1$): Customer opened many support tickets last month
- ($X_2 = 1$): Customer spent a lot of time on help / FAQ pages

Training dataset (historical sample)

Churn (Y)	$X_1 = 1$	$X_1 = 0$	Total
Y = 1	24	6	30
Y = 0	14	56	70

Churn (Y)	$X_2 = 1$	$X_2 = 0$	Total
Y = 1	25	5	30
Y = 0	13	57	70

(Note: each feature table is estimated independently, as in Naive Bayes.)

For an observed customer with: $X_1 = 1$, $X_2 = 1$, answer the following questions:

Question A1 — Estimate probabilities

Using **Laplace smoothing**, estimate:

- [5 pts] $P(X_1 = 1 \mid Y = 1)$, $P(X_1 = 1 \mid Y = 0)$
- [5 pts] $P(X_2 = 1 \mid Y = 1)$, $P(X_2 = 1 \mid Y = 0)$

Question A2 — Posterior probabilities

Using your smoothed estimates and Naive Bayes, compute the **normalized posterior probabilities**:

- [4 pts] $P(Y = 1 \mid X_1 = 1, X_2 = 1)$

- [4 pts] $P(Y = 0 \mid X_1 = 1, X_2 = 1)$

Question A3 — Prediction

Based on Model A:

- [1 pts] Does the model predict churn or no churn?
- [1 pts] What is the predicted probability?

Model B — Pricing and competition features

Feature definitions

- ($Z_1 = 1$): Customer changed pricing plan last month
- ($Z_2 = 1$): Customer received a competitor's marketing email

Training dataset (historical sample)

Churn (Y)	**Z_1 = 1**	**Z_1 = 0**	**Total**
Y = 1	18	12	30
Y = 0	21	49	70

Churn (Y)	**Z_2 = 1**	**Z_2 = 0**	**Total**
Y = 1	15	15	30
Y = 0	28	42	70

For an observed customer with: $Z_1 = 1$, $Z_2 = 1$, answer the following questions:

Question B1 — Estimate probabilities

Using **Laplace smoothing**, estimate:

- [5 pts] $P(Z_1 = 1 \mid Y = 1)$, $P(Z_1 = 1 \mid Y = 0)$
- [5 pts] $P(Z_2 = 1 \mid Y = 1)$, $P(Z_2 = 1 \mid Y = 0)$

Question B2 — Posterior probabilities

Using your smoothed estimates and Naive Bayes, compute the normalized posterior probabilities:

- [4 pts] $P(Y = 1 \mid Z_1 = 1, Z_2 = 1)$
- [4 pts] $P(Y = 0 \mid Z_1 = 1, Z_2 = 1)$

Question B3 — Prediction

Based on Model B:

- [1 pts] Does the model predict churn or no churn?
- [1 pts] What is the predicted probability?

Model comparison and interpretation

Compare the predictions from Model A and Model B:

- [5 pts] Why are the predicted churn probabilities so different?
- [5 pts] Which model's prediction is more trustworthy? Why?

Problem 2 [50 pts]:

Complete the Hidden Markov Model (HMM) Forward/Backtracking table (reference in published slides on HMM) as we have gone through in the class. Using these transition and emission probabilities.

Transition:

From → To	**Sunny**	**Cloudy**	**Rainy**
Sunny	0.33	0.67	0.00
Cloudy	0.33	0.00	0.67
Rainy	0.33	0.33	0.33

Emission:

Weather → Behavior	**Walk**	**Umbrella**
Sunny	4/4 = 1.0	0/3 = 0.0
Cloudy	2/3 ≈ 0.67	1/3 ≈ 0.33
Rainy	1/3 ≈ 0.33	2/3 ≈ 0.67

You can manually compute (show the steps), code (submit the code through github) or use a spreadsheet (submit the spreadsheet) to work out the solutions.